PR-1

Predictive Insight Engine

Supervised Learning

Q.1) What is Supervised Learning Algorithm?

→ Supervised learning algorithms learn from labelled data to predict outputs for new data.

Q.2) Regression Vs Classification Algorithm

→ Regression predicts numerical values, while classification predicts categories or classes

Q.3) Explain Simple Linear Regression.

→ It predicts an output using one input variable.
Its equation is $y = b_0 + b_1 x$,

Q.4) Assumption of Linear Regression.

→
- Linearity
- Independence of errors
- Constant variance of errors
- Normally distributed residuals
- Low multicollinearity (for multiple regression)

Q.5) What is Bias - Variance Trade - off?

→ It is the balance between bias and variance. High bias causes underfitting, while high variance causes overfitting.